

Activity 5

1. FileInputStream and FileOutputStream

A. Write and Read Single Data

```
import java.io.*;

public class SingleDataStream {

    public static void main(String[] args) throws IOException {

        // Write single byte

        FileOutputStream fos = new FileOutputStream("data.txt");

        fos.write(65); // ASCII value of A

        fos.close();

        // Read single byte

        FileInputStream fis = new FileInputStream("data.txt");

        int data = fis.read();

        System.out.println("Data read from file: " + (char)data);

        fis.close();

    }

}
```

Output

Data read from file: A

B. Write and Read Bulk Data

```
import java.io.*;

public class BulkDataStream {

    public static void main(String[] args) throws IOException {

        String text = "Welcome to Java File Handling";

        // Write bulk data

        FileOutputStream fos = new FileOutputStream("data.txt");

        byte[] bytes = text.getBytes();

        fos.write(bytes);

    }

}
```

```

        fos.close();

        // Read bulk data
        FileInputStream fis = new FileInputStream("data.txt");
        byte[] buffer = new byte[1024];
        int count = fis.read(buffer);
        String result = new String(buffer, 0, count);
        System.out.println("Data read from file: " + result);
        fis.close();
    }
}

```

Output

Data read from file: Welcome to Java File Handling

2. FileReader and FileWriter

A. Write and Read Single Character

```

import java.io.*;

public class SingleDataReaderWriter {

    public static void main(String[] args) throws IOException {

        // Write single character
        FileWriter fw = new FileWriter("data.txt");
        fw.write('A');
        fw.close();

        // Read single character
        FileReader fr = new FileReader("data.txt");
        int data = fr.read();

        System.out.println("Character read from file: " + (char)data);
        fr.close();
    }
}

```

Output

Character read from file: A

B. Write and Read Bulk Data

```
import java.io.*;

public class BulkDataReaderWriter {

    public static void main(String[] args) throws IOException {

        String text = "Java File Reader and Writer Example";

        // Write bulk characters

        FileWriter fw = new FileWriter("data.txt");

        fw.write(text);

        fw.close();

        // Read bulk characters

        FileReader fr = new FileReader("data.txt");

        char[] buffer = new char[1024];

        int count = fr.read(buffer);

        System.out.println("Data read from file:");

        System.out.println(new String(buffer, 0, count));

        fr.close();

    }

}
```

Output

Data read from file:

Java File Reader and Writer Example

3. Read Number of Lines and Number of Characters in a File

```
import java.io.*;

public class FileCount {

    public static void main(String[] args) throws IOException {

        FileReader fr = new FileReader("data.txt");

        int ch;

        int characters = 0;

        int lines = 0;

        boolean hasData = false;

        while ((ch = fr.read()) != -1) {

            characters++;

            hasData = true;

            if (ch == '\n') {

                lines++;

            }

        }

        // Handles the last line if it doesn't end with \n

        if (hasData) {

            lines++;

        }

        fr.close();

        System.out.println("Number of characters: " + characters);

        System.out.println("Number of lines: " + lines);

    }

}
```

Example data.txt

Hello Java

Welcome to Programming

File Handling Example

Output

Number of characters: 58

Number of lines: 3

4. Two Files Read Mode + One File Write Mode

Program

```
import java.io.*;

public class SequenceFileStream {

    public static void main(String[] args) throws IOException {

        // Open first file in read mode

        FileInputStream file1 =

            new FileInputStream("file1.txt");

        // Open second file in read mode

        FileInputStream file2 =

            new FileInputStream("file2.txt");

        // Combine two input streams

        SequenceInputStream sis =

            new SequenceInputStream(file1, file2);

        // Open third file in write mode

        FileOutputStream file3 =

            new FileOutputStream("output.txt");

        int data;

        // Read from both files and write to output file

        while ((data = sis.read()) != -1) {

            file3.write(data);

        }

        System.out.println("Files successfully combined.");
```

```
        sis.close();  
        file3.close();  
        file1.close();  
        file2.close();  
    }  
}
```

file1.txt

This is File 1.

file2.txt

This is File 2.

output.txt

This is File 1.

This is File 2.